

VeriDoc

Documentation Infrastructure Audit Report

Prepared by VeriOps

2026-06-01 11:53 UTC

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Risk: High

Key Metrics

Pages scanned: 1205

Report type: Premium Executive Audit

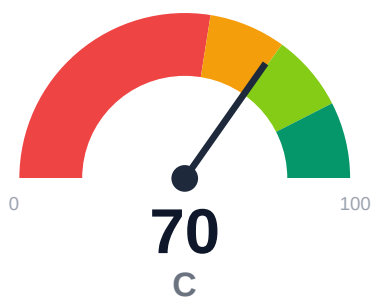
Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Crawl coverage is low: 6.53% (1337 URLs examined of 20474 in discovered scope). Run authen
- Broken internal links: 60
- Link verification inconclusive for 16497 URLs (auth/rate-limit/server block); these are ex
- API usage-doc coverage is low: 32.26% (73 API pages detected)

This report is generated automatically by the VeriOps platform. Estimates are directional.

Executive Summary



Documentation audit across 7 properties crawled 1,205 pages with 6.53% crawl coverage of 20,474 discovered in-scope pages. 60 confirmed broken internal documentation links identified. API reference coverage stands at 32.26% across 73 detected API pages. Code example reliability estimated at 16.95%, and only 3.4% of pages display freshness metadata. Overall audit confidence is 49.48% due to crawl limitations and 16,497 unverified links blocked by authentication or rate limiting.

External posture: SEO/GEO issue rate **7.5%**, public API coverage **32.3%**, broken links **60**.

70 Audit Score	C Grade	High Risk Band
1205 Pages Scanned	60 Broken Links	7.5% SEO Issue Rate

Per-Site Breakdown

Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://lightning.readthedocs.io/	1	1	N/A	0.0	75.0
https://wally.readthedocs.io/en/release_1.5.0	9	41	N/A	0.0	50.0
https://blockstream.com/satellite-api-document	34	1	100.0	0.0	26.5
https://docs.liquid.net/docs/building-on-liqu	34	3	33.3	97.8	42.6
https://gdk.readthedocs.io/en/release_0.76.3	31	14	N/A	0.0	50.0
https://help.blockstream.com/hc/en-us	171	0	29.8	100.0	26.6
https://github.com/Blockstream/docs/blob/mast	925	0	36.4	0.0	0.0

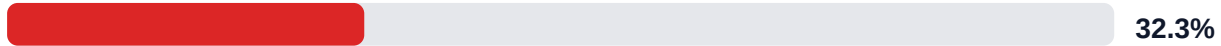
Industry Benchmark Comparison

Benchmark	Score	Gap
Top performers (Stripe, Twilio)	92+	-22
Industry average (developer docs)	85	-16
Minimum acceptable	70	-0
Your score	70	-

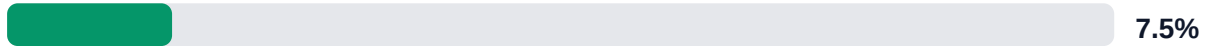
Gap to industry average: **16 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics

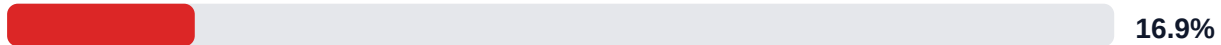
API Coverage (Public)



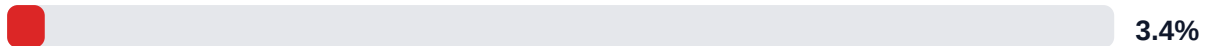
SEO/GEO Issue Rate (lower is better)



Example Reliability (estimate)

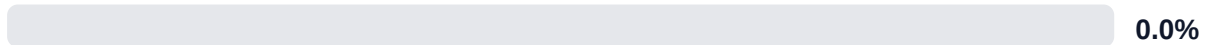


Freshness Visibility (last-updated metadata)



Internal Metrics (from scorecard)

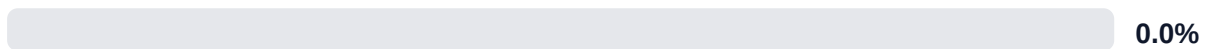
API Coverage (Internal)



Example Reliability (Internal)



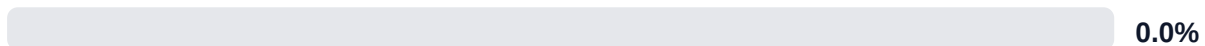
Docs-Contract Drift



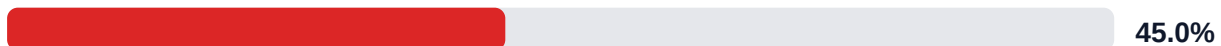
Pipeline Opportunity Fit (Deterministic)

This section maps cold-audit signals to problem classes that the pipeline can remediate directly. No LLM inference is used in this mapping.

LLM Retrieval Readiness (deterministic proxy)



Audit Confidence (crawl scope + verification quality)

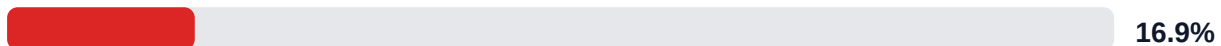


Readiness band: **Unknown**. Confidence: **Unknown**. Open remediation opportunities: **0**.

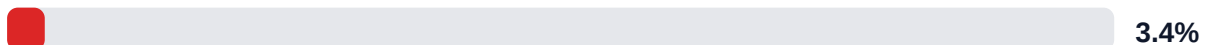
SEO/GEO Quality Signal (inverse issue rate)



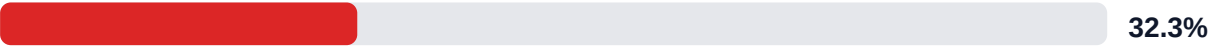
Example Reliability Signal



Freshness Visibility Signal



API Coverage Signal



Evidence coverage: **0.0%** (0/0 key claims backed by deterministic evidence signals).

Financial Exposure Model

Low Estimate \$90,406 annual exposure Remediation: \$1,852 Monthly loss: \$3,605 Opportunity: \$3,774/mo	Base Estimate \$185,064 annual exposure Remediation: \$4,798 Monthly loss: \$11,248 Opportunity: \$3,774/mo	High Estimate \$344,465 annual exposure Remediation: \$7,790 Monthly loss: \$24,282 Opportunity: \$3,774/mo
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Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-LAYERS	Missing required doc layers	HIGH	\$1,539	\$712
F-FRESHNESS	Documentation freshness debt	MEDIUM	\$969	\$475
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM	\$1,949	\$855
F-EVIDENCE-DRIFT-MISSING	Drift evidence report is missing	MEDIUM	\$104	\$95
F-API-COVERAGE	Undocumented API operations	LOW	\$2,508	\$950
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW	\$1,938	\$808
F-DRIFT	Code/docs contract drift	LOW	\$1,642	\$570
F-TERMINOLOGY	Terminology inconsistency	LOW	\$598	\$332

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-LAYERS	Missing required doc layers	HIGH
F-FRESHNESS	Documentation freshness debt	MEDIUM
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM
F-EVIDENCE-DRIFT-MISSING	Drift evidence report is missing	MEDIUM
F-API-COVERAGE	Undocumented API operations	LOW

Recommended Actions

1. Enable authenticated crawling and increase crawl budget to reach minimum 80% coverage of 20,474 discovered pages [impact: critical, effort: medium]
2. Remediate all 60 confirmed broken internal documentation links immediately [impact: critical, effort: low]
3. Implement automated example testing pipeline to improve 16.95% reliability score above 80% threshold [impact: high, effort: high]
4. Add last-updated metadata to all documentation pages (currently 3.4% coverage) using CI/CD automation [impact: high, effort: low]
5. Expand API reference documentation to achieve minimum 70% coverage from current 32.26% [impact: high, effort: high]

Expert Analysis: Strengths and Risks

Strengths

- + Multi-project topology correctly configured; no fragmentation penalty applied across 7 documentation properties
- + Zero broken repository navigation links (repo_broken_links_count: 0); all GitHub/GitLab internal links functional
- + API page detection successful with 73 pages identified and no detection failures flagged
- + SEO/geo issue rate contained at 7.45%
- + No trap URL patterns causing significant crawl waste (122 skipped)

Risks

- 60 confirmed broken internal documentation links degrading user navigation and trust
- Crawl coverage critically low at 6.53% (1,205 of 20,474 pages); 93.47% of documentation not audited
- 16,497 links unverified due to authentication/rate-limiting; actual broken link count potentially much higher
- API reference coverage at 32.26% leaves 67.74% of API surface undocumented or undetected
- Example reliability at 16.95% indicates majority of code samples may be outdated, untested, or non-functional
- Freshness metadata present on only 3.4% of pages; users cannot assess content currency
- Overall audit confidence at 49.48% limits reliability of all findings and metrics

Limitations

- * Crawl reached only 6.53% of discoverable documentation; findings do not represent 93.47% of content that was inaccessible or skipped
- * 16,497 links could not be verified due to authentication requirements, rate limiting, or server blocks; true broken link count unknown
- * External audit cannot validate technical accuracy, completeness of explanations, or correctness of architectural guidance
- * Example reliability is estimated through heuristics; actual runtime validity requires environment-specific testing
- * Multi-site topology may hide content duplication, versioning conflicts, or canonical URL issues not detectable through crawl analysis

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	95.0
support_hourly_usd	45.0
release_count_per_month	4.0
baseline_manual_sync_hours_per_week	8.0
avg_release_delay_hours	3.0
monthly_support_tickets	300.0
docs_related_ticket_share	0.25
avg_ticket_handling_minutes	18.0

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
F-EVIDENCE-DRIFT-MISSING -- Drift evidence report is missing	pr_docs_contract.json (missing)	MEDIUM
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	LOW
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	HIGH

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.